

STRIDE Phase II - Professional Estimation & Delivery Plan

Project: TMS V2 (STRIDE Phase II) **Organization:** Mazik Global, Inc. **Date:** June 12, 2026 **Timeline:** 90-100 Calendar Days (13-14 Weeks) **Status:** Ready for Execution

Executive Summary

STRIDE Phase II introduces 10 core features to the existing TMS platform, extending capabilities in task management, automation, collaboration, and mobile access. All features will be delivered within a 90-100 day execution window with UAT readiness and go-live support.

Total Effort: 165 person-days **Recommended Team:** 4 professionals (2 Backend, 1 Frontend, 1 QA) **Delivery Model:** Wave-based rollout (5 waves over 13-14 weeks) **Go-Live Date:** Week 14 (End of day 100)

Feature Scope (Locked)

#	Feature	V2 Inclusion	Effort (Days)
1	Task Follow-up	✓ Full	8
2	Private AI Engine (Local Model)	✓ Full	15
3	Meeting Notes (without AI)	✓ Full	10
4	Local Media Engine + Sharing + Recording	✓ MVP	18
5	External Sync (Local)	✓ Core	12
6	Personal Calendar	✓ Full	9
7	Custom Email Alerts	✓ Full	8
8	Physical Room Management	✓ Full	13
9	Meeting Notes (with AI)	✓ Full	8
10	Mobile Version	✓ MVP	38
TOTAL			139

Notes:

- ? Features 4 & 10 are scoped as Minimum Viable Product (MVP) for V2.0. Advanced enhancements scheduled for V2.1 (post-launch).
- ? Feature 2 uses on-premise local AI (Ollama) only. No cloud AI dependencies.
- ? All features include production hardening, multi-tenant security, and test coverage.

Team Structure (90-100 Days)

Core Squad (Full-Time, Weeks 1-14)

? **Backend Lead + Backend Developer** (2 FTE)

? Data models, API design, integrations, local AI gateway, database migrations

? **Frontend Developer** (1 FTE)

? Angular UI implementation, prototype-to-production, responsive design, Capacitor setup

? **QA Engineer** (1 FTE, starting Week 3)

? Test case design, regression testing, UAT coordination, production readiness

Stakeholder Roles

? **Product Owner** (you / team lead): Requirements clarification, UAT sign-off, go-live decision

? **DevOps / Infrastructure**: Local AI host setup, deployment pipeline, production monitoring

90-100 Day Delivery Timeline

Week 1-3: Foundation & Quick Wins (Wave 1)

Features: #1 Task Follow-up, #6 Personal Calendar, #7 Custom Email Alerts

Deliverables:

? Task Follow-up: Backend CRUD, reminder notifications, database schema

? Personal Calendar: Personal event model, recurrence engine, iCal import/export

? Email Alerts: Rule engine, template system, event hook integration

? All features: Unit tests, basic integration tests, API documentation

Go-Live Date for Wave 1: End of Week 3

Week 4-6: Collaboration Features (Wave 2)

Features: #3 Meeting Notes (without AI), #8 Physical Room Management

Deliverables:

? Meeting Notes: Rich-text editor, participant linking, search indexing

? Room Management: Room booking engine, availability calendar, conflict prevention

? Notifications: Real-time updates via SignalR

? API test coverage, permission boundary tests

Go-Live Date for Wave 2: End of Week 6

Week 7-9: Intelligence Layer (Wave 3)

Features: #2 Private AI Engine (Local), #9 Meeting Notes (with AI)

Deliverables:

- ? Private AI Engine: Ollama integration, IAIService abstraction, tenant-scoped config, usage logging
- ? AI Meeting Notes: Auto-summarize, action-item extraction, suggestion panel
- ? Local model hardening, timeout/fallback handling, performance tuning
- ? Full integration tests with mock model, tenant isolation validation

Go-Live Date for Wave 3: End of Week 9

Week 10-12: Integration & Synchronization (Wave 4)

Features: #5 External Sync (Local), #4 Media Engine (MVP)

Deliverables:

- ? External Sync: Provider abstraction (AD/Exchange/Files), scheduler, conflict resolution, audit log
- ? Media Engine (MVP): Chunked upload pipeline, streaming endpoints, secure sharing links, recording session tracking
- ? Performance tests, stress tests for media, sync conflict recovery tests

Go-Live Date for Wave 4: End of Week 12

Week 13-14: Mobile & Release Hardening (Wave 5)

Features: #10 Mobile Version (MVP), UAT, Go-Live Support

Deliverables:

- ? Mobile: Capacitor setup, responsive layouts (auth, dashboard, tasks, chat, notifications), push notification tokens
- ? Full system UAT in staging environment
- ? Production readiness review: security audit, performance baseline, monitoring alerts
- ? Deployment runbooks, rollback procedures, support documentation
- ? Go-Live execution with 24-hour support standby

Go-Live Date: End of Week 14 (Day 100)

Feature Details & Deliverables

1. Task Follow-up

- ? Task follow-up entity linked to tasks
- ? CRUD API endpoints with authorization
- ? Reminder trigger via background job
- ? SignalR notifications
- ? Database migration script

2. Private AI Engine (Local Model)

- ? IAIService interface abstraction
- ? OllamaAIService implementation
- ? Tenant-scoped AI configuration management
- ? Chat/Summarize/Extract endpoints
- ? Conversation history persistence
- ? Rate limiting & usage tracking
- ? No external API calls (on-premise only)

3. Meeting Notes (without AI)

- ? Meeting note entity with rich-text content
- ? Participant tagging and mentions
- ? Full-text search
- ? CRUD API with filtering by date/workspace/participants
- ? Participant notification system
- ? Export to PDF capability

4. Local Media Engine (MVP)

- ? Chunked file upload with resumption
- ? Metadata storage (duration, thumbnail, file type)
- ? Streaming endpoint with range-request support
- ? Secure sharing with signed temporary URLs
- ? Recording session entity for meetings
- ? Multi-tenant storage partitioning

5. External Sync (Local)

- ? Sync profile entity (source, credentials, interval)
- ? Provider pattern: Active Directory, Exchange Calendar, Local File System
- ? Scheduler service (IHostedService)
- ? Conflict detection & last-write-wins resolution

- ? Sync audit log
- ? Encrypted credential storage

6. Personal Calendar

- ? Personal event entity (user-scoped, not workspace-scoped)
- ? Recurrence rule support (RRULE)
- ? Reminder notifications
- ? iCalendar (ICS) import/export
- ? Calendar view (month, week, day)

7. Custom Email Alerts

- ? Email alert rule entity (trigger, conditions, template)
- ? Email template entity with variable placeholders
- ? Rule evaluator service (event-driven hooks)
- ? Tenant-scoped rule management endpoints
- ? Test email capability

8. Physical Room Management

- ? Room entity (name, capacity, amenities, location)
- ? Room booking entity (date/time, attendees, status)
- ? Availability check service (overlap detection)
- ? Recurring booking support
- ? Booking confirmation/cancellation notifications

9. Meeting Notes (with AI)

- ? Summarization endpoint (via local AI model)
- ? Action item extraction (creates tasks automatically)
- ? Suggestion panel (grammar, structure, missing items)
- ? Versioned summary history
- ? Depends on Features #2 and #3

10. Mobile Version (MVP)

- ? Capacitor framework setup for iOS/Android
- ? Responsive layouts: login, dashboard, task list, board view, chat, calendar, notifications
- ? Native plugins: camera, push notifications, biometric auth (optional phase 2)
- ? Offline queue (basic sync on reconnect)
- ? Device token management for push notifications
- ? App Store & Play Store build pipeline ready

Technical Requirements & Infrastructure

Backend

- ? .NET 8 Clean Architecture (existing patterns)
- ? Multi-tenant data isolation (tenant ID in all queries)
- ? Entity Framework Core with EF Migrations
- ? SignalR for real-time updates
- ? Background services for reminders/sync/media processing

Frontend

- ? Angular 17+ with existing component library
- ? Responsive design (mobile-first for Wave 5)
- ? Capacitor for mobile packaging
- ? State management (existing patterns)

Infrastructure

- ? Local AI hosting (Ollama or equivalent)
- ? Suggested models: llama3.1:8b (chat), mistral:7b (summarization)
- ? Estimated compute: GPU or CPU with 16GB+ RAM
- ? Data residency: On-premise only
- ? Media storage: Local disk or MinIO (tenant-partitioned)
- ? Database: SQL Server (existing)
- ? Deployment: IIS / Docker (existing)

Testing & Quality Assurance

Test Coverage by Wave

Wave	Unit Tests	Integration Tests	UAT Scope
1-4	80%+	Core paths	Partial (features only)
5	85%+	Full system	Complete (all features, cross-feature)

UAT Activities (Week 13)

- ? User acceptance testing on staging environment
- ? Cross-feature integration validation
- ? Performance baseline establishment

? Rollback & recovery drill

Go-Live Checklist (Week 14)

- ? All 10 features UAT-approved
- ? Database migration tested on production clone
- ? Performance targets met (response time, throughput)
- ? Monitoring & alerting configured
- ? Support team trained

Deployment Schedule

Phase	Start Date	End Date	Duration	Milestone
Wave 1	Week 1	Week 3	3 weeks	Features #1, #6, #7 live
Wave 2	Week 4	Week 6	3 weeks	Features #3, #8 live
Wave 3	Week 7	Week 9	3 weeks	Features #2, #9 live
Wave 4	Week 10	Week 12	3 weeks	Features #5, #4 (MVP) live
Wave 5	Week 13	Week 14	2 weeks	Feature #10 (MVP), UAT, Go-Live
Total	Week 1	Week 14	14 weeks	All features UAT-ready & live

Success Criteria

✓ All 10 features deployed to production by Day 100 ✓ UAT sign-off from product owner / stakeholders ✓ Zero critical bugs in post-launch review ✓ Performance benchmarks met (API response times, media streaming) ✓ All users can access and use core flows (no training delays) ✓ Support team trained and ready for production issues

Assumptions & Constraints

1. **Team Availability:** All 4 team members fully allocated for 14 weeks, no context switching.
2. **Scope Lock:** The 10 listed features are fixed. Out-of-scope requests go to V2.1.
3. **Local AI Infrastructure:** Ollama host provisioned and ready before Wave 3 starts.
4. **Existing Systems:** TMS v1 architecture, auth, database remain stable. No breaking changes to core APIs.
5. **UAT Environment:** Production-like staging environment available by Week 13.
6. **Daily Sync:** 15-minute standup, no meetings > 1 hour per day to maximize dev time.

Risk Mitigation

Risk	Likelihood	Mitigation
Local AI model latency	Medium	Benchmark models in Week 7, pre-cache responses, fallback to simple rules
Media upload failures on slow connections	Medium	Resume capability, retry logic, progressive upload feedback
Room booking conflicts	Low	Unit test all overlap scenarios, live conflict display
External sync source unavailability	Medium	Graceful degradation, sync log alerts, manual trigger option
Mobile release delays	Medium	Start App Store setup in Week 1, parallel to dev (no blockers)

Approval & Sign-Off

Date: June 12, 2026 **Document Status:** Final – Ready for Execution

Next Steps:

1. Confirm team assignment and availability
2. Provision local AI infrastructure (Ollama host)
3. Schedule Week 1 kickoff
4. Set up production staging environment for UAT
5. Begin Wave 1 development immediately upon approval